SDR AND HAM REFERENCE

Radio Band Chart

All the bands, and the frequency ranges they cover - from the full radio spectrum down to the exact channels you can tune on an RTL-SDR today. US (FCC) allocations. Keep it by the radio.

ELF-VLF	LF	MF	HF	VHF	UHF	SHF	EHF+
3 Hz - 30 kHz	30-300 kHz	0.3-3 MHz	3-30 MHz	30-300 MHz	0.3-3 GHz	3-30 GHz	30 GHz +

The whole spectrum: the 12 named bands

Every radio signal lives in one of these bands. Each band is ten times wider than the one before it. The band also gives the wavelength - HF waves are 100 to 10 metres long, which is why ham bands are named in metres.

Band	Frequency range	Wavelength	What lives there
ELF	3 - 30 Hz	100,000 - 10,000 km	Submarine communication research; natural Earth signals
SLF	30 - 300 Hz	10,000 - 1,000 km	Power-line frequencies, submarine communication
ULF	300 Hz - 3 kHz	1,000 - 100 km	Mine communication, geophysics
VLF	3 - 30 kHz	100 - 10 km	Time signals, navigation, submarines
LF	30 - 300 kHz	10 - 1 km	Longwave broadcast (outside the US), beacons, time signals
MF	300 kHz - 3 MHz	1 km - 100 m	**AM broadcast**, maritime, 160m ham band
HF	3 - 30 MHz	100 - 10 m	**Shortwave**, ham HF, CB radio, over-the-horizon radar
VHF	30 - 300 MHz	10 - 1 m	**FM broadcast, NOAA weather, airband**, 2m ham, TV
UHF	300 MHz - 3 GHz	1 m - 10 cm	TV, cell phones, 70cm ham, GMRS, Wi-Fi, GPS, ADS-B
SHF	3 - 30 GHz	10 - 1 cm	Wi-Fi 5 GHz, satellite, radar, 5G, microwave links
EHF	30 - 300 GHz	1 cm - 1 mm	5G mmWave, radio astronomy, high-speed links
THF	300 GHz - 3 THz	1 - 0.1 mm	Experimental; the border with infrared light

> **How to read a frequency:** 1 MHz = 1,000 kHz = 1,000,000 Hz. SDR++ shows frequency as groups, so `000.000.162.400` is 162.400 MHz - NOAA, while `1.600.000` in the last groups is 1.6 MHz, down in the AM broadcast band.

Broadcast bands

Service	Frequency range	Mode	Notes
AM broadcast (US)	530 - 1700 kHz	AM	10 kHz channel steps. Sports talk lives here - Omaha's 1620 The Zone is at 1.620 MHz.
Longwave (Europe/Asia)	153 - 279 kHz	AM	Not used for broadcast in the US.
FM broadcast (US)	88 - 108 MHz	WFM	Odd tenths only: 88.1, 88.3 ... 107.9. Bandwidth 200000 in SDR++.
TV - VHF low	54 - 88 MHz	Digital (ATSC)	Channels 2-6. Largely quiet in most cities now.
TV - VHF high	174 - 216 MHz	Digital (ATSC)	Channels 7-13.
TV - UHF	470 - 608 MHz	Digital (ATSC)	Channels 14-36 in current US use.

Shortwave broadcast bands (HF)

Shortwave stations bounce off the ionosphere, so these travel worldwide - best after dark for the lower bands, in daylight for the higher ones. Named by wavelength, like the ham bands.

Band	Frequency range	Band	Frequency range
120m	2.300 - 2.500 MHz	**25m**	11.600 - 12.100 MHz
90m	3.200 - 3.400 MHz	**22m**	13.570 - 13.870 MHz
75m	3.900 - 4.000 MHz	**19m**	15.100 - 15.800 MHz
60m	4.750 - 5.060 MHz	**16m**	17.480 - 17.900 MHz
49m	5.900 - 6.200 MHz	**13m**	21.450 - 21.850 MHz
41m	7.100 - 7.600 MHz	**11m**	25.670 - 26.100 MHz
31m	9.250 - 9.900 MHz	**Time signals: WWV / WWVH on 5.000, 10.000, 15.000, 20.000 MHz (AM/USB)**	

Bands you can hear today on the RTL-SDR

Service	Frequency range	Mode / BW	What you will hear
NOAA Weather Radio	162.400 - 162.550 MHz	NFM / 12500	Computer voice, forecasts 24/7. Seven channels: 162.400, .425, .450, .475, .500, .525, .550. Omaha is 162.400 (KIH61).
Railroad	160.215 - 161.565 MHz	NFM / 12500	Dispatchers and train crews, in bursts. The big spike near 160.55 MHz is this band.
Airband (aviation voice)	118.000 - 136.975 MHz	AM / 10000	Pilots and towers. 108-118 MHz just below it is navigation signals, not voice.
Marine VHF	156 - 162 MHz	NFM / 12500	Ships and Coast Guard. Channel 16, 156.800 MHz, is the distress/calling channel.
2m ham repeaters	144 - 148 MHz	NFM / 12500	Repeater outputs sit in 147-148 and parts of 146. 146.940 MHz is K0USA, Ak-Sar-Ben, in Omaha.
70cm ham / public safety	420 - 450 MHz	NFM / 12500	Ham repeaters and links. Public-safety voice also lives nearby at 450-470 MHz.

Listening ranges worth knowing

Range	Frequency	Mode	Notes
VHF low band	30 - 50 MHz	NFM	Older public safety, business, and 6m ham at the top end. Mostly quiet now.
VHF high band	150 - 174 MHz	NFM	The busy one: NOAA, railroad, marine, fire/EMS paging, business radio.
UHF public safety	450 - 470 MHz	NFM	Police/fire/business in many cities; also GMRS at 462/467 MHz.
700 / 800 MHz trunking	769-869 MHz	Digital	Most metro police/fire are here, usually P25 digital. SDR++ alone cannot decode P25 - it needs decoder software, and encrypted traffic stays encrypted.
ADS-B aircraft	1090 MHz	Data	Aircraft position broadcasts - what dump1090 and Flightradar24 decode. Not audio.
GPS L1	1575.42 MHz	Data	Satellite navigation timing signals. Far too weak and wide to listen to as audio.
ISS / amateur satellite	145.800 MHz downlink	NFM	Space Station voice and SSTV downlink; uplinks differ. Passes last minutes - timing is everything.

Amateur (ham) bands - HF

US ham bands, named by wavelength. Voice on HF is SSB: **LSB below 10 MHz, USB above 10 MHz** - the convention worth memorising for the Technician exam. Rows marked in red are where a Technician has HF privileges.

Band	Frequency range	Typical use
160m	1.800 - 2.000 MHz	LSB voice, CW. Night-time, regional.
80m	3.500 - 4.000 MHz	LSB voice. Technician: CW only, 3.525-3.600 MHz.
60m	Five channels near 5.330 MHz	USB voice, channelised - not a normal band. General class and up.
40m	7.000 - 7.300 MHz	LSB voice, CW, data. Technician: CW only, 7.025-7.125 MHz.
30m	10.100 - 10.150 MHz	CW and data only - no voice, by rule.
20m	14.000 - 14.350 MHz	USB voice. The main daytime DX band. General class and up.
17m	18.068 - 18.168 MHz	USB voice, CW. General class and up.
15m	21.000 - 21.450 MHz	USB voice. Technician: CW only, 21.025-21.200 MHz.
12m	24.890 - 24.990 MHz	USB voice, CW. General class and up.
10m	28.000 - 29.700 MHz	USB voice and FM at the top. Technician: CW/data 28.000-28.300, SSB voice 28.300-28.500 MHz.

Amateur bands - VHF, UHF and microwaves

A Technician holds full privileges on every band on this page - all modes, full power limits. This is why most new hams start on 2m and 70cm.

Band	Frequency range	Typical use
6m	50 - 54 MHz	SSB, FM, repeaters. "The magic band" - opens without warning.
2m	144 - 148 MHz	FM repeaters and simplex, SSB, satellites, APRS on 144.390 MHz.
1.25m	222 - 225 MHz	FM repeaters and SSB. (219-220 MHz is digital-only.)
70cm	420 - 450 MHz	FM repeaters, satellites, links.
33cm	902 - 928 MHz	FM, data, weak signal. Shared with ISM devices (and Meshtastic).
23cm	1240 - 1300 MHz	FM, ATV (amateur TV), weak signal.
13cm	2300-2310, 2390-2450 MHz	Weak signal, data, satellite.
Microwaves	3.3, 5.7, 10, 24 GHz and higher bands to 250 GHz	Contesting, weak signal, experiments.

License-free and everyday radio

Service	Frequency range	Rules in one line
CB radio	26.965 - 27.405 MHz	40 channels, AM and SSB. No license. Ch 9 = emergency, Ch 19 = highway.
MURS	151.820, 151.880, 151.940, 154.570, 154.600 MHz	5 VHF channels, NFM, 2 watts. No license.
FRS	462 / 467 MHz (22 channels)	Handhelds only, no license. Shares frequencies with GMRS.
GMRS	462 / 467 MHz	Higher power and repeaters. License required - covers the family, no exam.
Key fobs / remotes	315 MHz and 433 MHz	Short data bursts - they show up on a waterfall as tiny blips, not audio.
Wi-Fi / Bluetooth	2.400 - 2.4835 GHz	Wi-Fi channels 1-11 (US). Bluetooth and microwave ovens share this band.
Wi-Fi 5 / 6 GHz	5.150 - 5.895 GHz, 5.925 - 7.125 GHz	Faster, shorter range. 6 GHz is Wi-Fi 6E/7.
Cell phones	600 / 700 / 850 / 1900 MHz, 2.5 / 3.5 GHz, 24-39 GHz	Paired up/down links, carrier-dependent. 24 GHz and up is 5G mmWave.

The modes in SDR++ - which one, where

The mode is the demodulator: how the radio turns a signal back into sound. The wrong mode is the number-one cause of faint, hissy, or garbled audio on a signal that is actually strong.

Mode	Stands for	Typical bandwidth	Use it for
WFM	Wide FM	200000	FM broadcast radio, 88-108 MHz. Uncheck Stereo on weak stations - mono is far clearer.
NFM	Narrow FM	12500	NOAA, railroad, marine, public safety, ham repeaters, a handheld radio.
AM	Amplitude Modulation	10000	AM broadcast, and airband - pilots and towers, 118-137 MHz.
USB	Upper Sideband	2500 - 3000	HF ham voice above 10 MHz, shortwave utilities, military.
LSB	Lower Sideband	2500 - 3000	HF ham voice below 10 MHz (80m, 40m).
DSB	Double Sideband	6000	AM with the carrier removed. Rare in the wild - mostly experimental.
CW	Continuous Wave	500	Morse code. Ham CW segments, repeater ID beeps, beacons.
RAW	Raw IQ	-	No demodulation. Hands the raw signal to other software to decode. Not for listening.

Antenna length cheat sheet

A whip works best at about a quarter wavelength. Formula: **`length in inches = 2808 / frequency in MHz`** (that is 234 / MHz, in feet, times 12). A dipole uses one quarter-wave element on each side. An antenna cut for the wrong band is the classic reason a strong station sounds faint - a 2.4 GHz stub is roughly 15 times too short for NOAA or FM.

Target	Frequency	Quarter-wave length
FM broadcast	98 MHz (mid-band)	About 29 inches per element
Airband	127 MHz (mid-band)	About 22 inches
2m ham	146 MHz	About 19.5 inches
NOAA / railroad / marine	160 - 162 MHz	About 17 - 17.5 inches
70cm ham / GMRS	440 - 462 MHz	About 6 - 6.5 inches
CB radio	27 MHz	About 104 inches (8.7 ft) - why CB whips are so long
Wi-Fi / nRF24	2.4 GHz	About 1.2 inches - why that little stub is that size

> **Receiving is always legal; transmitting is where the license matters.** Listen anywhere on this chart. Transmit only where you are licensed (ham bands, with a callsign) or where no license is needed (CB, MURS, FRS). An RTL-SDR cannot transmit at all.

Allocations are US (FCC) and simplified for listening use - edges of some services vary by region, and ham privileges shown are the FCC Part 97 basics. For exam study, check the current ARRL band chart and the 2026-2030 Technician question pool.